

Literature Comparison Table

Paper	Approach	Dataset	Main contribution	Key takeaway
Zhang et al.	LSTM-RNN	Li-ion battery cells	Learns long-term capacity degradation dependencies	LSTM is suitable for sequential degradation data
Rouhi Ardeshiri & Ma	Multivariate GRU	NASA	Uses voltage, current and temperature-derived features	Multivariate features can improve RUL prediction
Zraibi et al.	DNN, RNN, GRU, LSTM	NASA + CALCE	Comparative study of deep-learning models	LSTM showed strong RUL performance
Bi et al.	LSTM vs TCN	NASA	Compared temporal architectures for SOH	TCN is a useful alternative to LSTM
TCN-LSTM study	TCN-LSTM, CNN-LSTM, TCN, LSTM	NASA + Oxford	Joint SOH and RUL estimation	Hybrid models can improve performance, but results vary by dataset
Temporal Transformer study	Transformer + Denoising Autoencoder	Battery datasets	Adds temporal encoding and attention	Operating time is important for RUL prediction
CNN-LSTM study	CNN-LSTM based architecture	NASA + Oxford	Feature extraction + temporal learning	Feature selection can improve prediction
ITCN study	Improved TCN	NASA	Multi-feature extraction + TCN	Feature quality strongly affects SOH estimation
PHM/deep-learning review	Multiple DL approaches	Multiple datasets	Reviews battery PHM approaches	Provides overall understanding of DL-based battery prognostics
Recent LOCO study	LSTM, TCN, TFT	NASA + Oxford	Controlled comparison using Leave-One-Cell-Out validation	Realistic cross-cell validation is important

Dataset Requirements

Fields needed to model battery degradation, grouped by category

Basic battery information

- Battery/Cell ID
- Cycle number
- Time/operating time

Electrical measurements

- Voltage
- Current
- Capacity

Thermal information

- Temperature

Preliminary Dataset Structure

The preliminary dataset structure is:

Cell ID	Cycle	Time	Voltage	Current	Temperature	Capacity	SOH	RUL
Cell 1	1	t_1	V_1	I_1	T_1	Q_1	SOH_1	RUL_1
Cell 1	2	t_2	V_2	I_2	T_2	Q_2	SOH_2	RUL_2
Cell 1	3	t_3	V_3	I_3	T_3	Q_3	SOH_3	RUL_3

Dataset Quality Risks

Missing values

Missing voltage, current, temperature or capacity measurements can affect model training.

Noise

Battery measurements may contain measurement noise that can affect degradation trends.

Outliers

Abnormal measurements can distort the learned degradation relationship.

Incomplete cycles

Incomplete charging/discharging cycles may affect feature extraction.

Capacity fluctuations

Battery capacity may not decrease perfectly monotonically. Therefore, every increase should not automatically be treated as an outlier.

Different operating conditions

Temperature and operating conditions can change the degradation behavior.

Cell-to-cell variability

Different cells can exhibit different degradation trajectories.

Consolidated Literature Findings

Finding 1 — Battery degradation is a temporal problem

Battery SOH and RUL change with cycle progression, making temporal models appropriate.

Finding 2 — Multiple deep-learning architectures are being used

LSTM, GRU, TCN, hybrid CNN/LSTM/TCN and Transformer-based approaches have all been investigated.

Finding 3 — Input-feature selection matters

Studies use capacity alone as well as voltage, current, temperature and engineered features.

Finding 4 — Dataset characteristics affect model performance

The same architecture does not necessarily perform best on every dataset.

Model Development Pipeline

Raw Battery Dataset



Data Cleaning



Feature Extraction



SOH / RUL Calculation



Train - Validation - Test Split



Data Preprocessing



LSTM / GRU / TCN Models



Model Training



Model Evaluation



SOH & RUL Prediction